MIL-B-55291C(MU)
28 September 1970
SUPERSEDING
MIL-B-55291B(MU)
11 April 1969

MILITARY SPECIFICATION

BATTERY, THERMAL

1. SCOPE

1.1 This specification covers one type of thermal battery.

2. APPLICABLE DOCUMENTS

2.1 Government documents. The following documents of the issue in effect on date of invitation for bids or request for proposal, form a part of this specification to the extent specified herein:

STANDARDS

MILITARY

MIL-STD-105 - Sampling Procedures and Tables for Inspection by Attributes.
MIL-STD-129 - Marking for Shipment and Storage.
MIL-STD-1168 - Lot Numbering of Ammunition.

DRAWINGS

US ARMY MUNITIONS COMMAND

EDGEWOOD ARSENAL

C14-15-922 - Pin Firing.
B14-15-935 - Thermal Battery.
IDL14-15-935 - Thermal Battery-Inspection Data List.

(Copies of specifications, standards, drawings and publications required by suppliers in connection with specific procurement functions should be obtained from the procuring activity or as directed by the contracting officer.)

2.2 Other publications. The following document forms a part of this specification to the extent specified herein. Unless otherwise indicated, the issue in effect on date of invitation for bids or request for proposal shall apply.

FSC 6135

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UNIFORM CLASSIFICATION COMMITTEE

Uniform Freight Classification

(Application for copies of these ratings, rules and regulations should be addressed to the Uniform Classification Committee, 202 Union Station, 516 West Jackson Boulevard, Chicago, Illinois 60606.)

3. REQUIREMENTS

3.1 Materials and components. All materials and components shall conform to Drawing B14-15-935. Materials and components not definitely specified shall be of an accepted commercial type and grade suitable for the purpose intended.

3.2 Assembly. The thermal battery shall be as shown on Drawing B14-15-935.

3.3 Case leakage. The thermal battery shall not emit bubbles after it has been subjected to a pressure of 80 pounds per square inch (psig) of nitrogen (N_2) for 4 hours after which within 10 seconds it shall be placed in a water bath containing a wetting agent at 160° ± 10° F for a minimum period of 30 seconds. The thermal battery shall be tested in accordance with 4.4.4.1.

3.4 Resistance. The resistance between the center terminal and case--prior to firing primer--shall be infinity on the RX1000 scale ohmmeter at 70° ± 10° F when tested in accordance with 4.4.4.2.

3.5 Operating characteristics. The operating characteristics of the thermal battery shall be determined under a 2 ohm load condition.

3.5.1 Voltage. The voltage of the thermal battery measured across its terminals during the current drain shall be a minimum of 1.3 volts when tested in accordance with 4.4.4.3.

3.5.2 Initiation. The thermal battery shall be initiated upon the application of 45 ± 5 inch-ounces of energy to the primer when tested in accordance with 4.4.4.3.

3.5.3 Delay time. The delay time shall be 0.3 to 0.75 second when tested in accordance with 4.4.4.3. The delay time is the interval, measured to the nearest 0.01 second, between initiation (see 3.5.2) and the automatic recording of any output voltage.

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3.5.4 Discharge life. The discharge life of the thermal battery shall be a minimum of 0.5 second during the interval the output voltage attains and exceeds 1.3 volts when tested in accordance with 4.4.4.3.

3.6 Environmental.

3.6.1 Extreme low temperature. The thermal battery shall be subjected to a temperature of -65° F for a period of 4 hours minimum and shall comply with the requirements of 3.5 while at -65° F when tested in accordance with 4.3.4.1.

3.6.2 Extreme high temperature. The thermal battery shall be subjected to a temperature of +165° F for a period of 4 hours minimum and shall comply with the requirements of 3.5 while at +165° F when tested in accordance with 4.3.4.1.

3.6.3 Temperature cycling. The thermal battery shall meet the requirements of 3.3, 3.4 and 3.5 after being subjected to three temperature cycles when tested in accordance with 4.3.4.1. One cycle consists of 4 hours minimum at -65° F followed by 4 hours minimum at +160° F. A maximum delay of 5 minutes is permitted between the transfer of the battery from one temperature extreme to the other.

3.6.4 High temperature storage. The thermal battery shall meet the requirements of 3.3, 3.4 and 3.5 after being subjected to a temperature of 160° F for a minimum period of 180 hours when tested in accordance with 4.3.4.1.

3.6.5 Transportation vibration. The thermal battery shall comply with the requirements of 3.3, 3.4 and 3.5 subsequent to the completion of the vibration criteria in table I when tested in accordance with 4.3.4.2.

3.6.6 Transportation shock. The thermal battery shall comply with the requirements of 3.3, 3.4 and 3.5 subsequent to being shocked at 75 ± 8 g peak with the shock having a time duration of 9 ± 3 milliseconds at half amplitude. The maximum time from initation of shock to peak g shall not exceed 7 milliseconds. The thermal battery shall be shocked in each of three mutually perpendicular directions. One of these shock directions shall be parallel to the cylindrical axis with the thermal battery dropped base down. All these operations shall be conducted in accordance with 4.3.4.3.

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Table I. Vibration

Frequency range cps	Double displacement or peak g level	Cycle time to cover frequency range (minutes)	Total test time in minutes at given temperature		
			80° ± 20°	160° ± 5°F	-65° ± 5°F
10-20-10	± 2 g	3 ± 0.5	30	10	10
20-75-20	0.036 inches	3 ± 0.5	15	10	10
75-1000-75	± 10 g	3 ± 0.5	15	10	10

3.7 Preproduction. Prior to the start of regular production, a preproduction lot of batteries shall be produced in accordance with this specification for examination and test (see 4.3).

3.8 Workmanship. The battery shall be free from foreign matter (oil, grease or dirt) and damage such as cracks, dents or deformation which could affect its intended use.

4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection.

4.1.1 Supplier's responsibility. Unless otherwise specified in the contract or purchase order, the supplier is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified in the contract or order, the supplier may use his own or any other facilities suitable for the performance of the inspection requirements specified herein, unless disapproved by the Government. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to assure supplies and services conform to specified requirements.

4.1.2 Objective evidence. The supplier shall provide objective evidence acceptable to the contracting officer that the requirements of 3.1 and section 5 for which specific inspection has not been provided in this specification have been satisfied.

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4.2 Classification of inspections.

(a) Preproduction inspection (see 4.3).
(b) Quality conformance inspection (see 4.4).

4.3 Preproduction inspection.

4.3.1 Lot. A preproduction lot of 1,216 batteries shall be manufactured using the same methods, materials, equipment and processes as will be used during regular production.

4.3.2 Sampling.

4.3.2.1 For examination and nondestructive tests. Sampling shall be conducted in accordance with MIL-STD-105, inspection level III.

4.3.2.2 For destructive tests. Sampling shall be conducted in accordance with table II.

4.3.3 Inspection procedure.

4.3.3.1 For examination and nondestructive tests. The sample batteries shall be examined and tested in accordance with the classification of defects (4.4.3.4). The AQL's do not apply.

4.3.3.2 For destructive tests. The sample batteries shall be tested in accordance with table II.

4.3.4 Tests.

4.3.4.1 Environmental. The battery shall be subjected to the specified temperatures for the required period of time and then functioned as required.

4.3.4.2 Transportation vibration. The battery shall be subjected to the specified temperature for the required period of time (see 6.3) and functioned as required.

4.3.4.3 Transportation shock. The battery shall be shocked as required for the specified period of time and then functioned as required.

4.3.5 Acceptance/rejection criteria. The preproduction samples shall comply with the examinations and tests specified in 4.3.3 to be acceptable. The acceptance number is zero. The supplier shall obtain written approval from the contracting officer prior to proceeding with regular production.

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Table II. Preproduction sampling and tests

Quantity	Requirement	Test sequence	Test
	3.6.5	Transportation vibration	4.3.4.2
20	3.3	Case leakage	4.4.4.1
	3.4	Resistance	4.4.4.2
	3.5	Functioning	4.4.4.3
	3.6.6	Transportation shock	4.3.4.3
20	3.3	Case leakage	4.4.4.1
	3.4	Resistance	4.4.4.2
	3.5	Functioning	4.4.4.3
	3.6.3	Temperature cycling	4.3.4.1
20	3.3	Case leakage	4.4.4.1
	3.4	Resistance	4.4.4.2
	3.5	Functioning	4.4.4.3
	3.6.2	High temperature	4.3.4.1
20	3.4	Resistance	4.4.4.2
	3.5	Functioning	4.4.4.3
	3.6.1	Low temperature	4.3.4.1
20	3.4	Resistance	4.4.4.2
	3.5	Functioning	4.4.4.3
	3.6.4	Storage	4.3.4.1
	3.3	Case leakage	4.4.4.1
200	3.4	Resistance	4.4.4.2
	3.5	Functioning	4.4.4.3

4.4 Quality conformance inspection.

4.4.1 Lotting. Each lot shall be identified and controlled in accordance with MIL-STD-1168. Contained in any one interfix number of batteries, no more than one interfix number from one manufacturer of the following component is permitted.

(a) Primer

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4.4.2 Sampling.

4.4.2.1 For examination and nondestructive tests. Sampling shall be conducted in accordance with MIL-STD-105.

4.4.2.2 For destructive tests. Sampling shall be conducted in accordance with MIL-STD-105, inspection level S-4.

4.4.3 Inspection procedure.

4.4.3.1 For examination and nondestructive tests. The sample batteries shall be examined and tested in accordance with the classification of defects and MIL-STD-105.

4.4.3.2 For destructive tests. The sample batteries shall be tested in accordance with 4.4.4.3 and MIL-STD-105 using an AQL of 1.5 percent defective.

4.4.3.3 For critical defects. Each item in the lot shall be tested for the critical characteristic listed in the classification of defects.

4.4.3.4 Classification of defects. (See inspection data list IDL 14-15-935 for inspection equipment required to perform the examinations specified in this section.)

(a) Battery, thermal (Drawing B14-15-935).

Categories	Defects	Acceptance standards
Critical:		
1	Resistance	4.4.4.2
Major:	AQL 1.0 percent defective	
101	Length of battery excessive	
102	Diameter of battery excessive	
103	Insulation diameter or length incorrect	
104	Seal diameter or length incorrect	
105	Terminal wire length incorrect	
	AQL 0.65 percent defective	
106	Case leakage	4.4.4.1

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4.4.4 Tests. (See inspection data list IDL 14-15-935 for inspection equipment required to perform the tests specified in this section.)

4.4.4.1 Case leakage. The thermal battery shall be immersed in the nitrogen bath at the required pressure for the required period of time. Within 10 seconds after removal from the nitrogen, the thermal battery shall be immersed in the water at the specified temperature for the required period of time.

4.4.4.2 Resistance. The center terminal and the case shall be connected to the leads of an ohmmeter and the resistance between the two leads recorded.

4.4.4.3 Operating characteristics. The thermal battery shall be tested as follows:

(a) Connect the thermal battery to the test circuit as specified.

(b) Drop a ball weighing between 4 and 8 ounces from the height required to exert the specified energy onto the firing pin (Drawing C14-15-922) resting on the primer of the thermal battery. The fall of the weight may be free or guided through a tube.

(c) Automatic recording equipment shall be used to record the required operating characteristics.

5. PREPARATION FOR DELIVERY

5.1 Packaging, level C. The batteries shall be packaged to provide adequate protection against corrosion, deterioration, and damage from the supply source to the first receiving activity for immediate use or for controlled humidity storage.

5.2 Packing, level C. Batteries, packaged as specified in 5.1, shall be packed to insure carrier acceptance and safe delivery to the first domestic destination. Containers shall be in compliance with Uniform Freight Classification rules or the regulations of other common carriers applicable to the mode of transportation.

5.3 Marking. In addition to any special marking required by the contract or order, unit packages and shipping containers shall be marked in accordance with MIL-STD-129.

6. NOTES

6.1 Intended use. This item is used as a component of a bomb fuze.

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6.2 Ordering data. Procurement documents should specify the following:

(a) Title, number, and date of this specification.
(b) Preproduction.

(1) Time allowed for supplier submission of samples for Government test and evaluation after award of contract.
(2) Name and address of test facility when testing is performed by the Government.
(3) Packing and shipping instructions when applicable.
(4) Time required for the Government to notify the supplier whether or not to proceed with production.

6.3 Vibrator. The NOL Type 2 vibrator is considered suitable vibrating equipment. Information regarding this vibrator may be obtained from the Naval Ordnance Laboratory, Silver Springs, Maryland.

Custodian:

Army - MU

Preparing activity:

Army - MU (EA)

Project No. 6135-A112

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